CLUSTERING ANALYSIS – STUDENT PERFORMANCE DATASET

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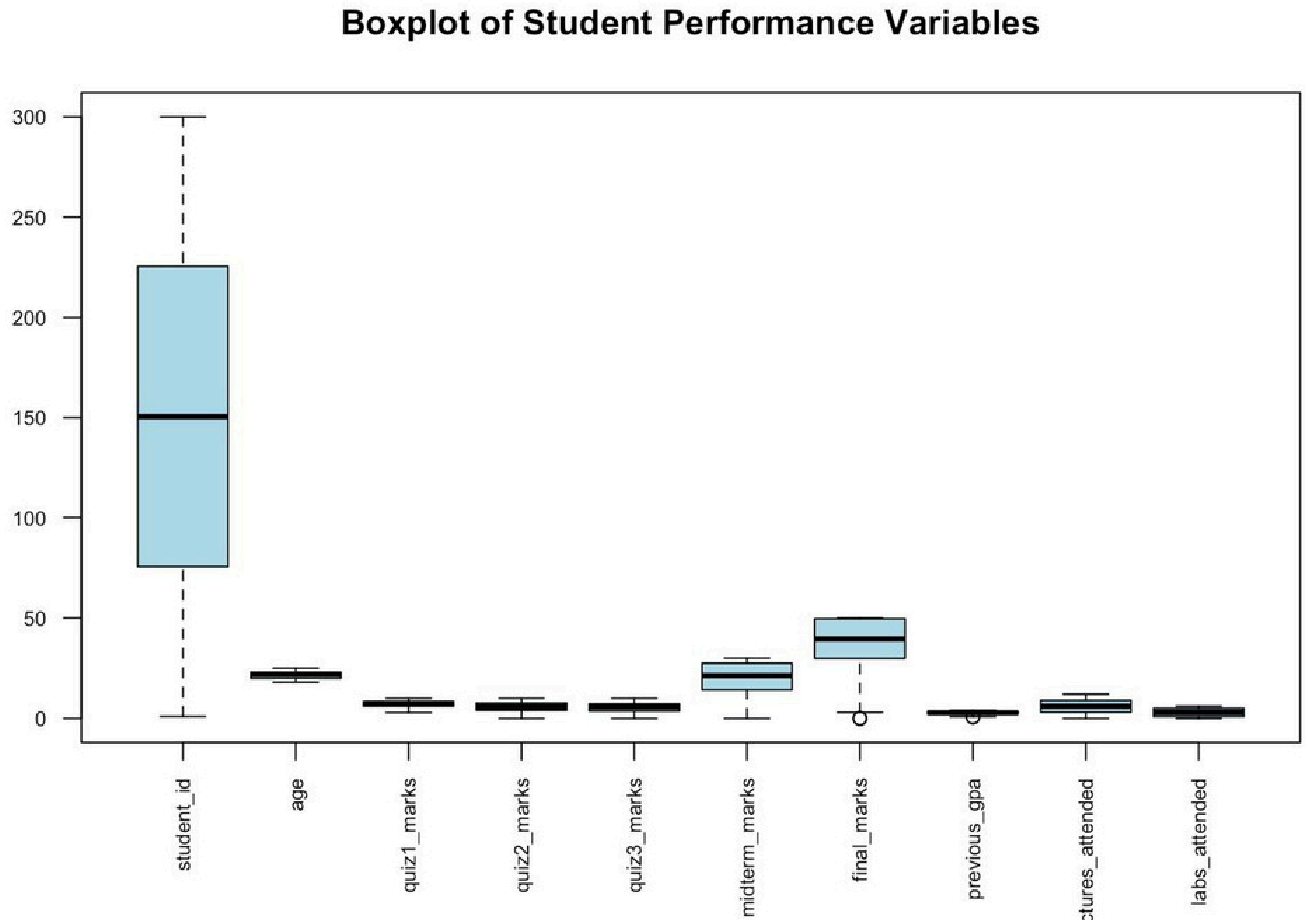
DATASET DESCRIPTION

- Source: Kaggle (muhammadkhubaibahmad/student-performance-and-clustering-dataset)
- Type: Student performance dataset
- Dimensions: 300 rows × 16 columns
- Data type: Mostly numeric scores for different subjects and assessments

```
> head(student_df, 10)     # First 10 observations
# A tibble: 10 × 16
   student_id name             age gender quiz1_marks quiz2_marks quiz3_marks total_assignments
        <dbl> <chr>          <dbl> <chr>        <dbl>       <dbl>       <dbl>             <dbl>
 1          1 Kristina Va…      19 Male           8           5.7         7.4                 5
 2          2 Rodney Dani…      21 Male          10           7.9         4.1                 5
 3          3 Jose Nash         19 Female         7.5         1.2         0.3                 5
 4          4 Nicole Mart…      21 Male           5.2         2.5         9.9                 5
 5          5 Shelby Smith      21 Female         5.9         6.3         2                   5
 6          6 Austin Grif…      22 Male           5.8        10           5.5                 5
 7          7 Crystal Jor…      23 Female         7.4         1.1         1.5                 5
 8          8 Sandra Patt…      23 Male           6.8         5.2         1.1                 5
 9          9 Rachael Mil…      23 Female         7.7         1.6         6.5                 5
10         10 Matthew Rice      18 Female         9.1         8.3         3                   5
```

DATA CLEANING AND PREPARATION



Variable Selection and Cleaning (Script Step 2)

The raw dataset was pre-processed for clustering:

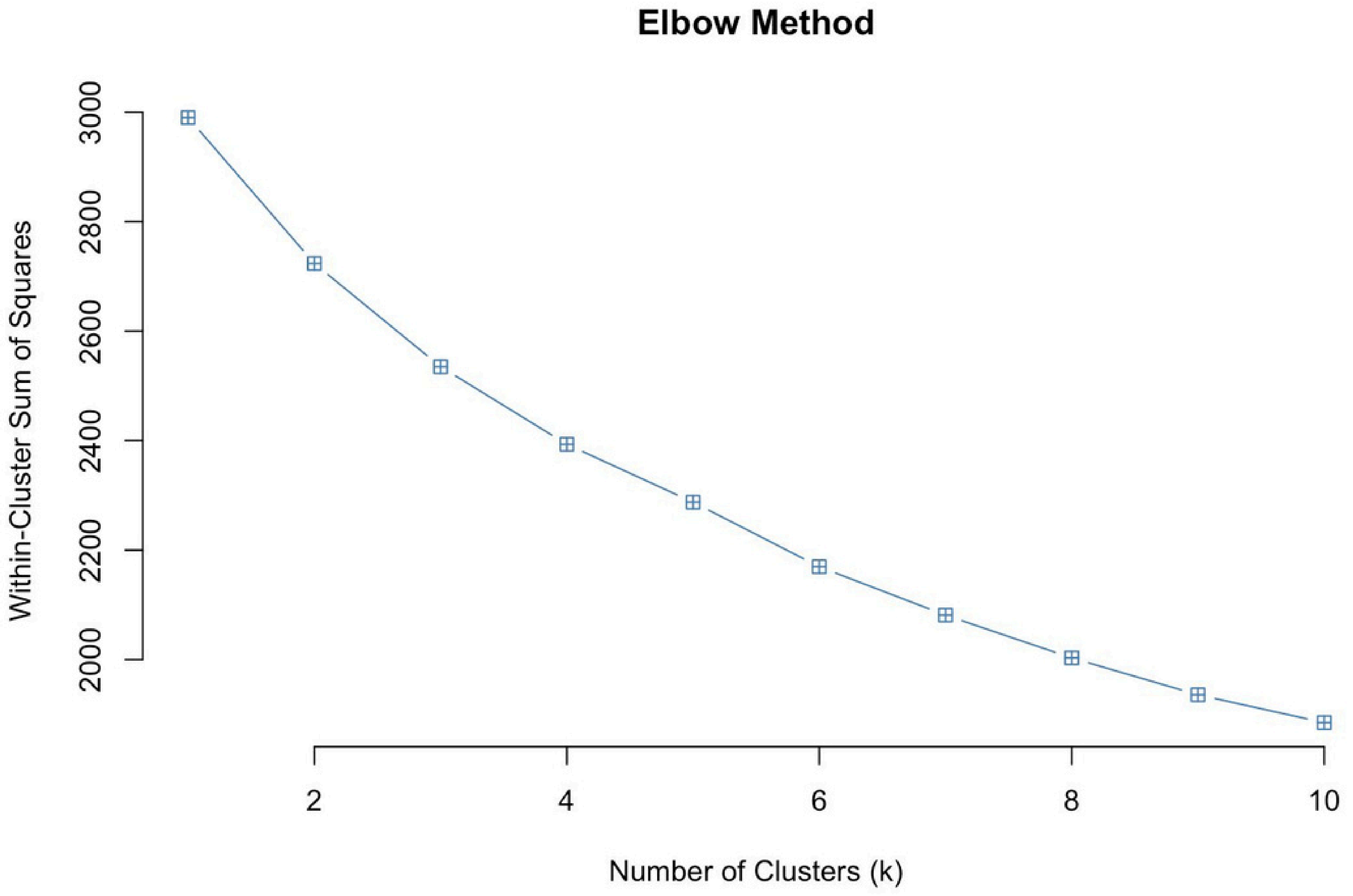
• Only **numerical variables** were retained.

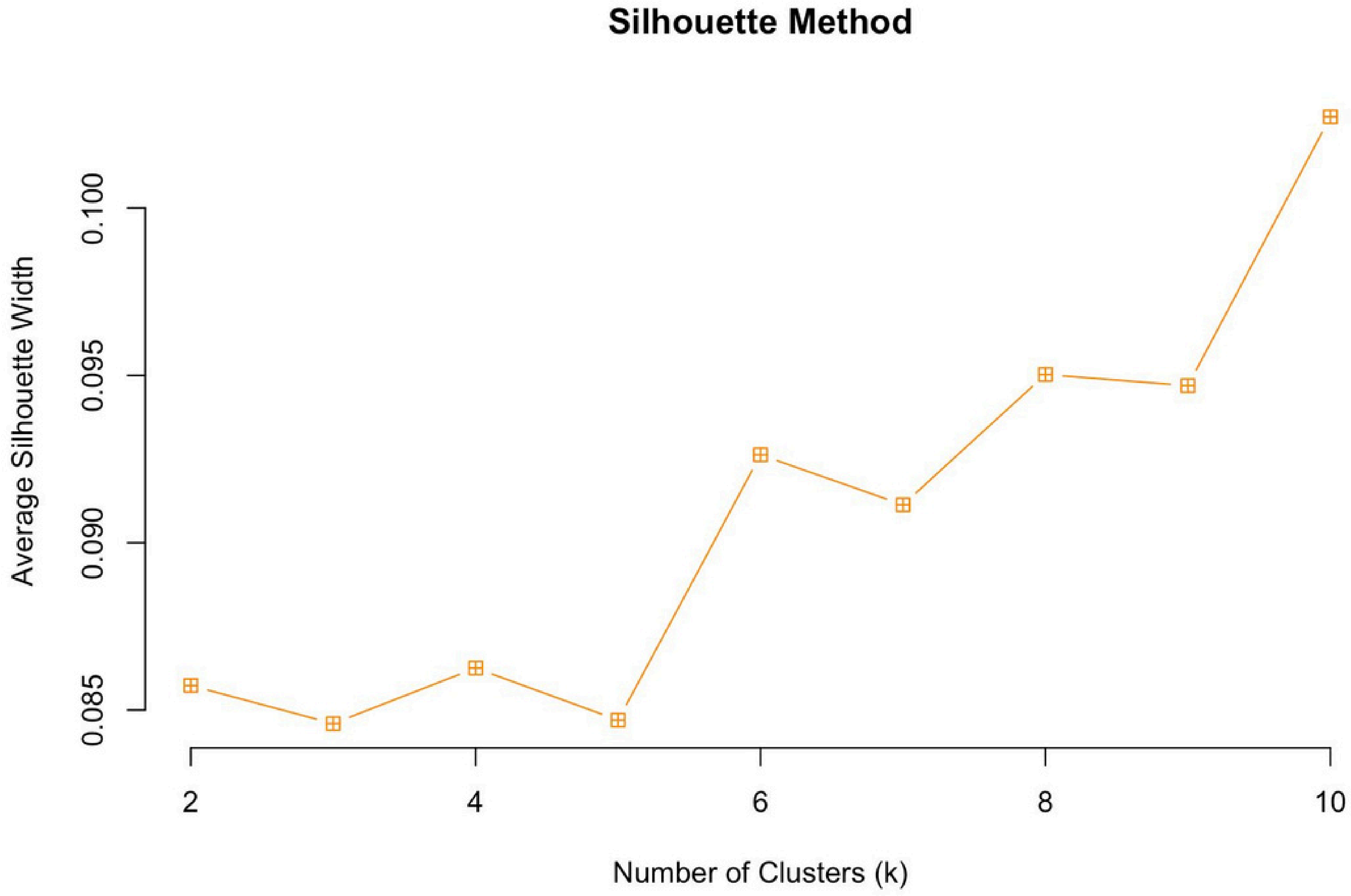
• Remaining **missing values** were imputed using the **median** of the respective column.

• Variables with **zero variance** (e.g., constants) were removed.

Determining Optimal Number of Clusters

- Optimal number of clusters: k = 3



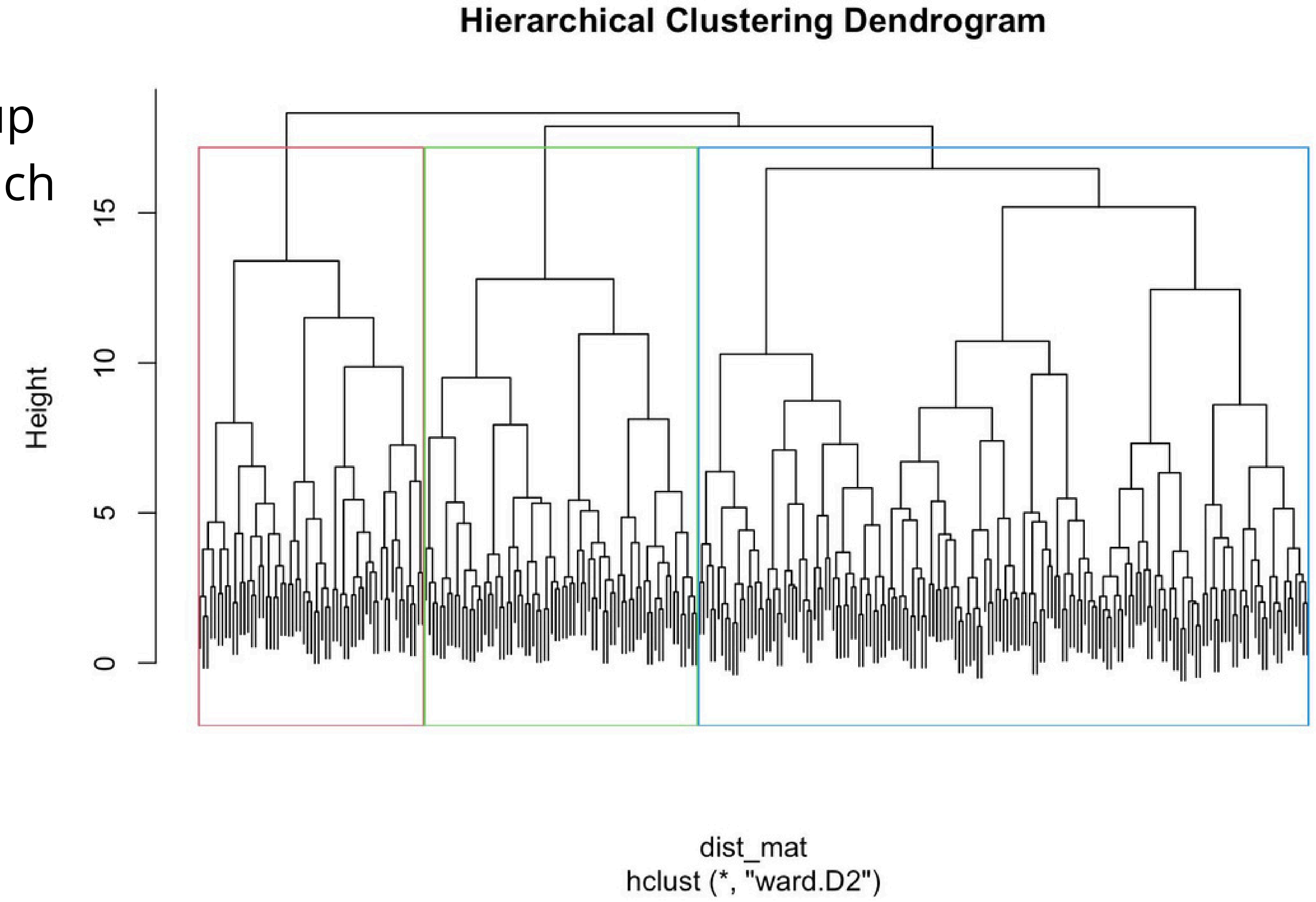


HIERARCHICAL CLUSTERING

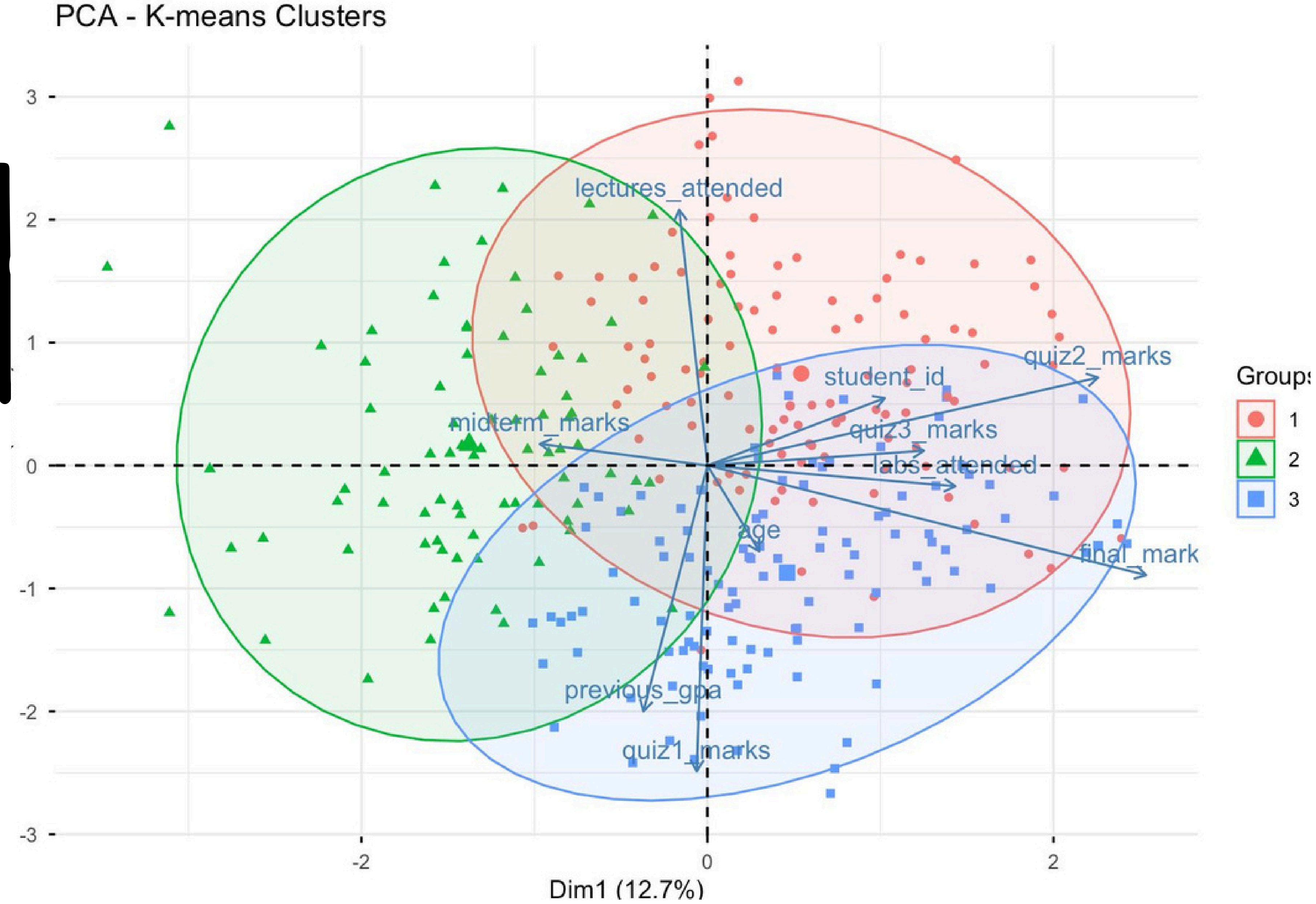
- Hierarchical clustering divided students into 3 clusters.

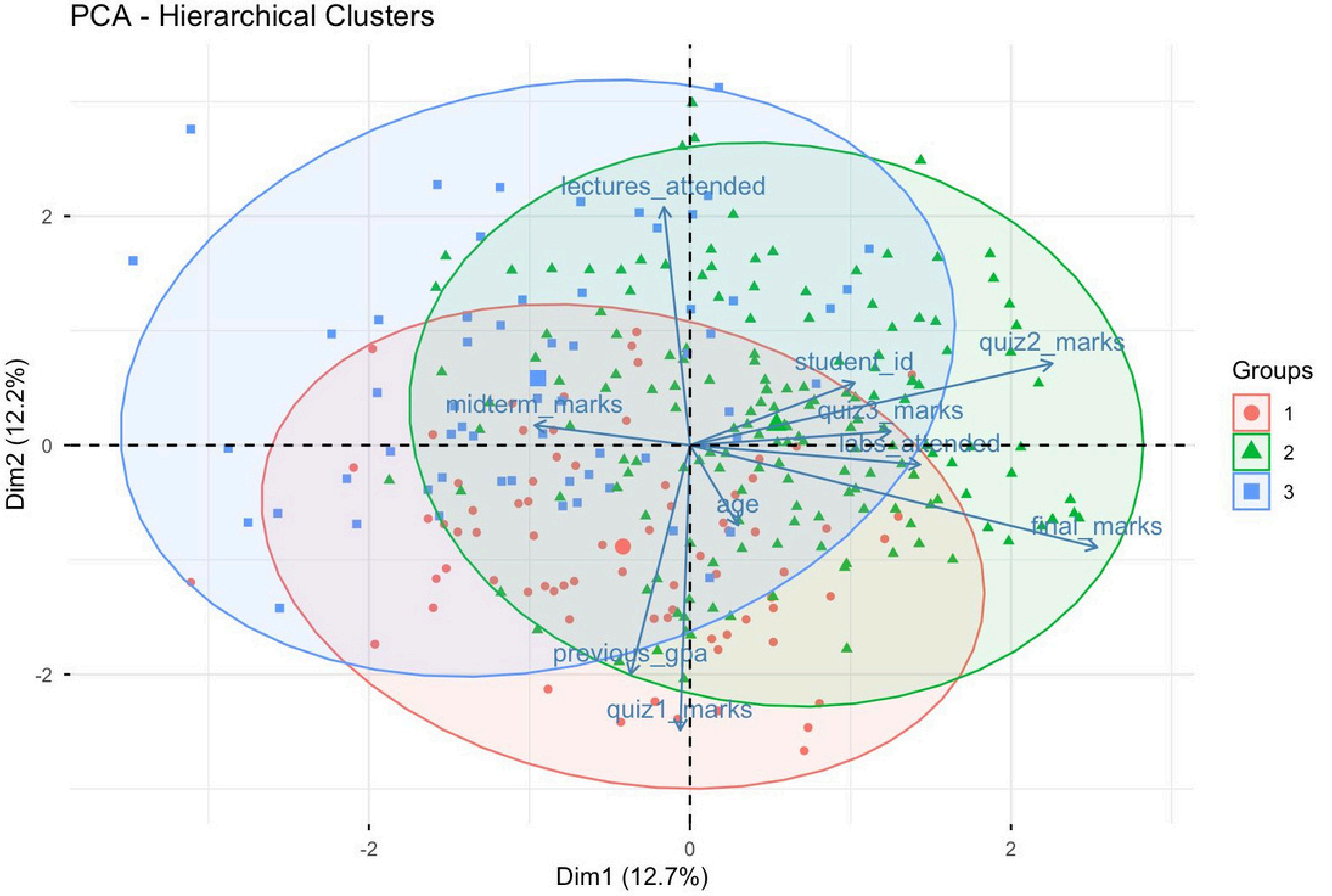
```
> table(hc_clusters)
hc_clusters
  1   2   3
 74 165  61
```

1. The dendrogram illustrates how students group hierarchically based on performance, with branch heights showing cluster dissimilarities.



PCA VISUALIZATION





PCA visualization shows the student distribution in the reduced two-dimensional space.

CLUSTERING SUMMARY

1. A printed table showing the mean values of each variable per K-means cluster.

```
> print("Cluster Summary (K-means):")
[1] "Cluster Summary (K-means):"
> print(cluster_summary)
# A tibble: 3 × 11
  Cluster_KM student_id     age quiz1_marks quiz2_marks quiz3_marks midterm_marks
  <fct>           <dbl>   <dbl>       <dbl>       <dbl>       <dbl>         <dbl>
1 1             0.0771   0.0507      -0.452     0.407         0.175       -0.0783
2 2            -0.115   -0.0948       0.0956   -0.551        -0.452        0.253
3 3             0.00618  0.0183       0.382    -0.00615       0.154       -0.106
```

Cluster summary shows average performance for each K-means group. Cluster 1 represents high-performing students, Cluster 2 medium, and Cluster 3 lower-performing students.

CONSLUSION

Have been analyzed the Student Performance Dataset and identified 3 clusters of students using K-means and Hierarchical clustering. Cluster 1 shows high-performing students, Cluster 2 medium, and Cluster 3 lower-performing. The Adjusted Rand Index **(0.872)** confirms strong agreement between methods. This analysis helps understand student performance patterns and can guide targeted support or interventions.